Low-Level Design (LLD)

RAG-Based Customer Support Assistant

1. Objective

The Low-Level Design (LLD) defines the internal implementation details of the RAG system, including modules, data structures, workflow logic, and execution flow.

2. Module-Level Design

1. Document Processing Module

- Loads PDF using PyPDFLoader
- Extracts raw text

2. Chunking Module

- Splits text into smaller chunks
- Chunk size: ~800 characters
- Overlap: ~150 characters

3. Embedding Module

- Converts text chunks into vectors
- Uses HuggingFace embedding model

4. Vector Storage Module

- Stores embeddings in ChromaDB
- Enables fast similarity search

5. Retrieval Module

- Accepts user query
- Converts query to embedding
- Retrieves top-k relevant chunks

6. Query Processing Module

- Combines retrieved chunks
- Sends context + query to LLM

7. Graph Execution Module

- Implements workflow using LangGraph
- Controls node execution

8. HITL Module

- Handles human intervention
- Activated for low-confidence queries

3. Data Structures

Document Representation

- Raw text extracted from PDF

Chunk Format

```
{
  "content": "text chunk",
  "metadata": "source info"
}
```

Embedding Structure

```
[0.123, 0.456, 0.789, ...]
```

Query-Response Schema

```
{
  "query": "user input",
  "answer": "generated response",
  "confidence": 0.75
}
```

Graph State Object

```
{
  "query": "",
  "answer": "",
  "confidence": 0.0
}
```

4. Workflow Design (LangGraph)

Nodes

- Processing Node
- Output Node

Edges

- Input → Processing Node
- Processing Node → Output Node

State

- Stores query, answer, confidence
- Passed between nodes

5. Conditional Routing Logic

Routing Conditions

- If confidence ≥ 0.3 → Output
- If confidence < 0.3 → HITL

Escalation Criteria

- No relevant chunks found
- Low similarity score
- Complex query

6. HITL Design

Trigger Condition

- Low confidence score

Process

1. System detects low confidence
2. Prompts human for input
3. Human response is used as final answer

Integration

- Human answer replaces AI-generated response
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7. API / Interface Design

Input Format

```
User query (string)
```

Output Format

```
Generated answer (string)
```

Interaction Flow

1. User enters query
2. System processes query
3. Response is displayed

8. Error Handling

Missing PDF

- Display error message

No Relevant Chunks

- Return fallback response

LLM Failure

- Retry or return default message

Conclusion

The LLD provides a detailed blueprint for implementing the RAG system. It ensures modularity, scalability, and efficient handling of user queries using structured workflows and conditional logic.